Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Federated Terrain-Adaptive Learning & Control for 4WS Agricultural Autonomous Mobile Robots**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		*Mechanical · Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Software	*Embedded Software*	*Robotics/Automation Machinery*	*Robot Control and Intelligence Technologies*
Concept of Proposed Technology	**Core Concept** This technology develops an integrated AI platform for agricultural autonomous mobile robots (AMRs) that optimizes both perception and control through advanced machine learning. The system employs federated learning (FL) for privacy-preserving collaborative perception across distributed farm sites, enabling crop/obstacle detection models to be trained without sharing raw farm data. Simultaneously, reinforcement learning (RL) enables terrain-adaptive steering control that automatically selects optimal driving modes (crab, diagonal, zero-turn) based on real-time terrain conditions. The platform is designed for bandwidth-constrained rural environments with gradient compression and asynchronous aggregation.			

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Rationale for Joint R&D Project	**Critical Technical Barriers** Korean agriculture faces critical challenges including aging farmer population, labor shortages, and the need for 24/7 autonomous operation in diverse field conditions. Korean SME partners have developed innovative hardware platforms with advanced steering technology but lack the AI algorithms needed for privacy-preserving learning across customer farms and adaptive control on variable agricultural terrain. **Why Global Collaboration?** Research institute provides complementary expertise in federated learning algorithms, reinforcement learning control, agricultural edge computing infrastructure, and instrumented vehicle datasets for simulation validation. This collaboration enables research to be directly validated on real agricultural robot deployments.
Key Words	1. Federated Learning 2. Reinforcement Learning 3. Agricultural Robotics 4. Terrain Adaptation 5. Autonomous Navigation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	**R&D Objectives** 1) Federated Perception Learning ● Develop bandwidth-efficient FL algorithms for on-farm edge training, enabling collaborative crop row detection, obstacle recognition, and terrain classification models without sharing raw farm data. 2) Terrain-Adaptive RL Control ● Build physics-validated simulation using instrumented agricultural vehicle terrain interaction data and develop RL controller for 4-wheel steering mode selection with sim-to-real transfer. 3) Integrated Autonomy Stack ● Couple FL perception outputs with RL control inputs and deploy staged autonomy (shadow → limited → full) on agricultural robot fleets with safety constraints. 4) Edge Computing Infrastructure ● Integrate edge computing nodes with cloud platforms for model aggregation, designed for rural connectivity limitations. 5) Field Validation ● Validate on Korean smart farms with measurable performance improvements in navigation success rate and operational efficiency.			
R&D Target Specifications	● FL Recognition Accuracy: High-precision segmentation under diverse conditions ● Data Transmission Reduction: Significant bandwidth minimization via Edge computing ● Terrain Navigation Success: Reliable autonomous navigation in unstructured environments ● Steering Mode Optimization: Autonomous real-time adaptation ● Model Deployment Time: Rapid over-the-air updates			
R&D Plan	**Phase 1: Foundation** 1) Definition of FL protocol architecture and edge-cloud interface 2) Collection of terrain interaction log data from instrumented vehicles 3) Development of FL perception prototype and terrain simulation environment 4) Design and validation of initial RL controller using partitioned datasets **Phase 2: Development & Integration** 1) Deployment of edge computing nodes at test farms 2) Training of first FL perception model using real-world farm data 3) Completion of Sim-to-Real transfer learning for RL control 4) Shadow mode-based integrated stack operation and validation			

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	Phase 3: Field Validation 1) Expansion of FL network to multiple farms 2) Phased deployment of autonomy (limited → full) 3) Comparative field trials against baseline control 4) Demonstration of robust navigation success **Phase 4: Commercialization** 1) Commercial integration into Partner's cloud management platform 2) Performance optimization and documentation 3) Knowledge transfer and training for Partner engineers 4) Production-ready deployment

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	● Lead the design and development of federated learning algorithms with gradient compression ● Develop terrain-adaptive reinforcement learning controller with sim-to-real transfer ● Provide instrumented vehicle datasets for simulation validation ● Design edge computing architecture for rural connectivity constraints ● Conduct algorithm validation using agricultural testbed infrastructure ● Publish research findings in top-tier robotics and AI venues ● Provide graduate student researchers and computational resources
Expected R&R of Korean SME Partner	● Provide access to 4-wheel steering agricultural robot platform ● Supply terrain sensor hardware and steering control interfaces ● Deploy edge computing nodes at partner farm sites in Korea ● Conduct field validation on Korean smart farms ● Integrate FL/RL algorithms into commercial cloud management platform ● Lead commercialization, manufacturing, and distribution ● Provide access to customer farms for real-world validation

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical Achievements** 1) First Federated Learning system for agricultural robot perception 2) Terrain-adaptive control trained on real-world agricultural data 3) Field-ready cloud-edge-robot autonomous pipeline **Intellectual Property** 1) Joint patent applications for core technologies 2) Publications in top-tier robotics and AI venues **Commercialization Potential** 1) Premium AI incentives for collaborative farms Autonomous terrain adaptation features for agricultural robots 2) Technology licensing for global agricultural robotics **Industrial/Economic Impact** 1) Advancement of agricultural autonomy through real-world validation 2) Cost reduction via reliable 24/7 robot operation 3) Foundation for data-driven smart farming ecosystem 4) Job creation in AI/Robotics R&D sectors

6. References

(1) McMahan, B., Moore, E., Ramage, D., Hampson, S., & y Arcas, B. A. (2017, April). Communication-efficient learning of deep networks from decentralized data. In Artificial intelligence and statistics (pp. 1273-1282). PMLR.

(2) Kairouz, P., McMahan, H. B., Avent, B., Bellet, A., Bennis, M., Bhagoji, A. N., ... & Zhao, S. (2021). Advances and open problems in federated learning. Foundations and trends® in machine learning, 14(1–2), 1-210.

(3) Dembani, R., Karvelas, I., Akbar, N. A., Rizou, S., Tegolo, D., & Fountas, S. (2025). Agricultural data privacy and federated learning: A review of challenges and opportunities. Computers and Electronics in Agriculture, 232, 110048.

(4) Yu, P., Teng, F., Zhu, W., Shen, C., Chen, Z., & Song, J. (2025). Cloud–edge–device collaborative computing in smart agriculture: architectures, applications, and future perspectives. Frontiers in Plant Science, 16, 1668545.

(5) Lee, J., Bjelonic, M., Reske, A., Wellhausen, L., Miki, T., & Hutter, M. (2024). Learning robust autonomous navigation and locomotion for wheeled-legged robots. Science Robotics, 9(89), eadi9641.

(6) Liu, G., Xiao, W., Han, X., Song, K., & Xie, H. (2024, October). Reinforcement Learning Optimized MPC for Vehicle Stability with Independent Four-Wheel Drive and Steering. In 2024 8th CAA International Conference on Vehicular Control and Intelligence (CVCI) (pp. 1-6). IEEE.

(7) Liu, G., Xiao, W., Han, X., Song, K., & Xie, H. (2024, October). Reinforcement Learning Optimized MPC for Vehicle Stability with Independent Four-Wheel Drive and Steering. In 2024 8th CAA International Conference on Vehicular Control and Intelligence (CVCI) (pp. 1-6). IEEE.

(max 10 pages)

(8) Peng, X. B., Andrychowicz, M., Zaremba, W., & Abbeel, P. (2018, May). Sim-to-real transfer of robotic control with dynamics randomization. In 2018 IEEE international conference on robotics and automation (ICRA) (pp. 3803-3810). IEEE.

(9) de Silva, R., Cielniak, G., & Gao, J. (2023). A Vision-Based Navigation System for Arable Fields. arXiv preprint arXiv:2309.11989.

(10) Qu, J., Zhang, Z., Qin, Z., Guo, K., & Li, D. (2024). Applications of autonomous navigation technologies for unmanned agricultural tractors: A review. Machines, 12(4), 218.

(11) Rajendran, V., Debnath, B., Mghames, S., Mandil, W., Parsa, S., Parsons, S., & Ghalamzan-E, A. (2024). Towards autonomous selective harvesting: A review of robot perception, robot design, motion planning and control. Journal of Field Robotics, 41(7), 2247-2279.